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A real world evaluation of an innovative artificial intelligence tool for population-level breast cancer screening



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In resource-constrained countries like India, mammography-based breast screening is challenging to implement. This state-wide study, funded by the Government of Punjab, evaluated the use of Thermalytix, a low-cost, radiation-free AI tool, for breast cancer screening. Community health workers, trained to raise awareness, mobilized women aged 30 and above for screening. Thermalytix triaged women into five risk categories based on thermal images, with high-risk women recalled for diagnostic imaging. Over 18 months, 15,069 women were screened across 183 locations in Punjab. The median age was 41 years, and 69.9% were asymptomatic. Of 460 women testing positive (recall rate 3.1%), 268 underwent follow-up imaging, and 27 were confirmed with breast cancer, yielding a detection rate of 0.18%. The positive predictive value of biopsy performed was 81.81%, and the median diagnostic interval was 21 days, with therapy initiation within 30 days. The study demonstrates the potential of Thermalytix for effective population-level breast cancer screening in low-resource settings.

Globally, breast cancer is a major cause of morbidity and mortality. In low- and middle-income countries (LMICs), changing lifestyle and reproductive factors among women including early menarche, delayed childbirth, reduced parity, and increasing body weight, are projected to increase breast cancer incidence in the upcoming years¹. In India, over 26 years from 1990 to 2016, the age-adjusted incidence (AAI) rate of breast cancer increased by 39.1%, with an observed rise in every state of the country².

In the state of Punjab, four population-based cancer registries have been established across different districts—Sangrur, Mansa, SAS Nagar, and Patiala—covering urban, semi-urban, and rural populations. These registries highlight the growing burden of cancer in the region, with breast cancer cases rising from 39,521 in 2021 to 42,288 in 2024, reflecting a 7% increase³. Additionally, the registries report that breast cancer has become the most common cancer among women, accounting for 30% of all cancer cases, with an age-adjusted incidence rate of 37.5 per 100,000, one of the highest rates in India. An 11% annual increase in breast cancer incidence was recently reported by the Indian Council of Medical Research, based on population-based cancer registries (PBCRs) in Punjab⁴. However, the actual number of new cases is likely to be higher, since the coverage of PBCRs is only 16.4%, and the rural component of PBCRs is not reported in most states⁵. Remarkably, participation rates in the existing breast cancer screening programs across India are as low as 0.9%, with participation in breast cancer

screening in Punjab itself being 0.3%, significantly below the national average⁶.

The World Health Organization (WHO) recommends population-based mammographic screening only in high-resource settings with well-funded, coordinated health systems. However, in low- and middle-income countries (LMICs), the absence of sustainable financing, cultural barriers, and limited availability of medical equipment and healthcare personnel make systematic mammographic screening challenging to implement. When launching the Global Breast Cancer Initiative in 2021⁷, the WHO addressed a common misconception that mammographic screening is essential to achieve the downstaging necessary for significant reductions in breast cancer mortality. The WHO emphasized that any screening program incorporating community awareness, primary care-based screening, enhanced referrals, and tracking systems to ensure timely diagnosis and treatment could be feasible and effective in LMICs. Consequently, the WHO advocates for a combination of clinical history, clinical breast examination (CBE), and diagnostic breast ultrasound in resource-constrained countries⁸. This framework underpins India's National Programme for Prevention and Control of Cardiovascular Disease, Diabetes, Cancer, and Stroke (NPCDCS), launched in 2016 (<http://www.mohfw.gov.in/?q=Major-Programmes/non-communicable-diseases-injury-trauma/Non-Communicable-Disease-II/National-Programme-for-Prevention-and->

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Control-of-Cancer-Diabetes-Cardiovascular-diseases-and-Stroke-NPCDCS).

To evaluate the effectiveness of CBE as a breast cancer screening modality in India, two high-quality cluster-randomized trials were conducted over the last decade. Both trials followed a similar protocol, comparing CBE screening with a no-intervention control group, and yielded mixed results. In the trial by Mittra et al.⁹, CBE significantly downstaged breast cancer at diagnosis across all age groups. However, a reduction in mortality was observed only in women aged 50 and above, with no such benefit in younger women. Conversely, the trial by Ramadas et al.¹⁰ did not report a statistically significant reduction in mortality, although a trend toward disease downstaging and improved survival in the intervention group was noted.

Moreover, a recent overview of systematic reviews found no consistent evidence supporting CBE's effectiveness in downstaging disease or reducing mortality rates¹¹. The conflicting evidence regarding its role in reducing mortality, coupled with the subjective nature of CBE, highlights the uncertainty surrounding its utility in breast cancer screening. To address screening gaps in LMICs, there is growing interest in developing novel screening tools that are not only clinically effective but also accessible, affordable, and acceptable, without relying heavily on highly skilled healthcare workers¹².

Artificial intelligence (AI) tools have emerged as transformative technologies in healthcare, with the potential to reduce healthcare disparities¹³. These tools offer enhanced accuracy, improved accessibility, optimized resource allocation, reduced workload, and cost-effectiveness. Globally, numerous AI-based solutions have been developed to address various healthcare needs. In breast imaging specifically, tools like ScreenPoint Medical's Transpara¹⁴ and CureMetrix's cmAssist¹⁵ claim to assist radiologists by increasing breast cancer detection rates in mammograms by over 25% and reducing mammogram reading time by 26%¹⁶. Similarly, Qlarity Imaging's QuantX AI algorithms aid radiologists in assessing and characterizing breast abnormalities, improving cancer detection from MRI images by over 35%¹⁷. Additionally, BUCAD supports the identification of suspicious regions in breast ultrasound images¹⁸.

In line with other AI-based tools, Thermalytix is a novel breast screening technique that uses advanced machine learning algorithms to analyze minute temperature variations in breast thermal scans, generating a quantitative report that indicates the likelihood of malignancy. This software-based medical device has received regulatory approval in India and holds the European CE mark. It is noncontact, non-invasive, radiation-free, and does not require breast compression, making it highly acceptable to patients. Furthermore, it is lightweight, portable, and requires minimal infrastructure, which enhances its feasibility in diverse healthcare settings. This makes Thermalytix a promising alternative to clinical breast examinations (CBE), particularly in resource-limited settings^{19–30}.

Over the past few years, several prospective and retrospective clinical studies have demonstrated the clinical effectiveness of Thermalytix in breast cancer detection. In 2016, Madu et al.¹⁹ published the results of the Thermalytix prototype, which showed the feasibility of machine learning-based breast cancer detection from thermal images. A pilot study in 2018 involving 147 patients reported 98% sensitivity in detecting breast cancers with Thermalytix²⁰. In 2020 a retrospective study of 470 women found a sensitivity of 91.02% (89.85% for symptomatic cases and 100% for asymptomatic cases) and a specificity of 82.39% (69.04% for symptomatic cases and 92.41% for asymptomatic cases)²¹. In a prospective study, conducted between 2017 and 2019 on 258 symptomatic women, demonstrated a sensitivity of 82.5% and a specificity of 80.5%²². Another prospective study, conducted in 2022 involving 459 women with both dense and fatty breast tissue, reported a sensitivity of 95.24% (100% for dense breasts) and a specificity of 88.58% (81.65% for dense breasts)²³. Additionally, community-level studies conducted in 2022 confirmed the feasibility of implementing Thermalytix screening in large-scale populations²⁴. These validation studies, conducted in various clinical settings, provided sufficient evidence for the Government of Punjab to initiate a new large-scale population field study with Thermalytix.

The main purpose of this study was to evaluate the effectiveness of Thermalytix for population-level breast cancer screening compared to the CBE, which is currently the standard breast screening modality in resource-constrained settings. The Department of Health and Family Welfare (DHFV) of the Government of Punjab initiated a public-private partnership with Niramai Health Analytix, m16 labs, and Roche India to improve breast cancer screening services in Punjab. Niramai, the developer of the innovative Thermalytix technology, was responsible for conducting the screening test, m16 labs provided a digital referral pathway, and Roche India was responsible for capacity building among existing health workers and helping create awareness among the public. This collaboration was set up to simultaneously address all three pillars of the WHO Global Breast Cancer Initiative⁷: health promotion and early detection, timely diagnosis, and comprehensive breast cancer management. In this paper, we present the initial 18-month experience of this field-based project, detailing the implementation and outcomes of this collaborative effort towards improving breast screening services in Punjab.

Results

Screening rate and geographic distribution

From 15th June 2022 to 23rd December 2023, 15,069 women were screened using Thermalytix across all 23 districts of Punjab, at 183 locations including all 23 district hospitals, 39 of 41 (95% coverage) sub-district hospitals, 104 of 161 (64.6% coverage) community health centres, 8 primary health centres, and in 9 isolated camps at schools, etc. (Fig. 1).

Out of 477 days available excluding Sundays, screening was performed on 443 days (92.8% utilization rate). This utilization allowed screening of 800–1000 women per month which likely reflects the increasing familiarity with the clinical pathway by HCWs as well as increasing acceptance of the program from the women being invited for screening (Fig. 2).

Demographics of the women screened

The median age of screened women was 41 (IQR 16 years), with 78.3% younger than 50. The majority were premenopausal (64.6%) and asymptomatic (69.9%). Demographic data are presented in Table 1.

Thermalytix screening outcomes and follow-up

Based on the calculated likelihood of malignancy, Thermalytix triages women in five risk categories, with high-risk women recommended for further diagnostic imaging. Out of the 15,069 women screened, Thermalytix triaged 460 for further diagnostic imaging. Of these, 455 women (98.9%) were contacted by the Patient Navigators (PNs) either via telephone or personal visits. Follow-up for 5 women was not possible due to incorrect phone numbers. Despite the PNs' best efforts, 187 participants (41.1%) refused to share their diagnostic reports or disclose whether they underwent diagnostic tests. Of these 187 participants, 130 (69.5%) women were symptomatic and the remaining 57 women did not report any symptoms. The remaining 268 women provided information about their follow-up diagnostic tests (Fig. 3).

Out of these 268 women, imaging reports were available for 150 women, out of whom 64 had a radiological finding and 33 underwent histological confirmation. 27 breast malignancies were confirmed histopathologically. The mean age in these women was 54.6 years (SD = 14.1). This translates into a crude incidence of 183.21 per 100,000 (AAI of 120.94/100,000) for women above the age of 30 (Supplementary Table 1). The overall recall rate (test positivity) was 3.1% (460/15,069). The recall rate for symptomatic women was 7.8% (352/4538), and 1% (108/10,531) in asymptomatic women. The positive predictive value in symptomatic and asymptomatic participants who completed follow-up was 11.3% and 4.3%, respectively. In women with a suspicion of breast lump ($n = 1557$), Thermalytix was negative in 1253 women and the remaining 304 women had a B-Score of 4 or 5 and were sent for diagnostic follow-up.

In asymptomatic women, 2 malignancies were detected in 30 out of 47 screened-positive women who had a follow-up imaging test; an additional 5

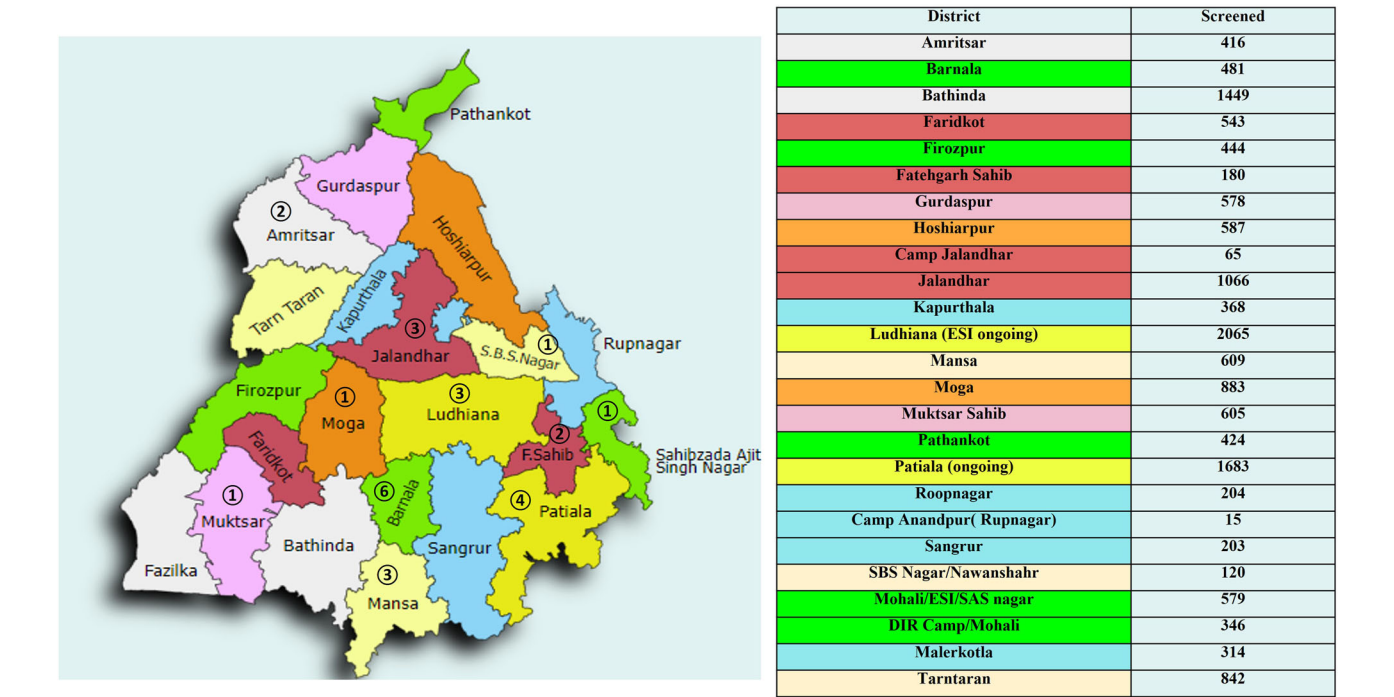


Fig. 1 | Distribution of screenings across Punjab. The number of screenings using Thermalytix across all 23 districts of Punjab. The numbers in the O indicate the number of confirmed breast cancer cases found in the respective district.

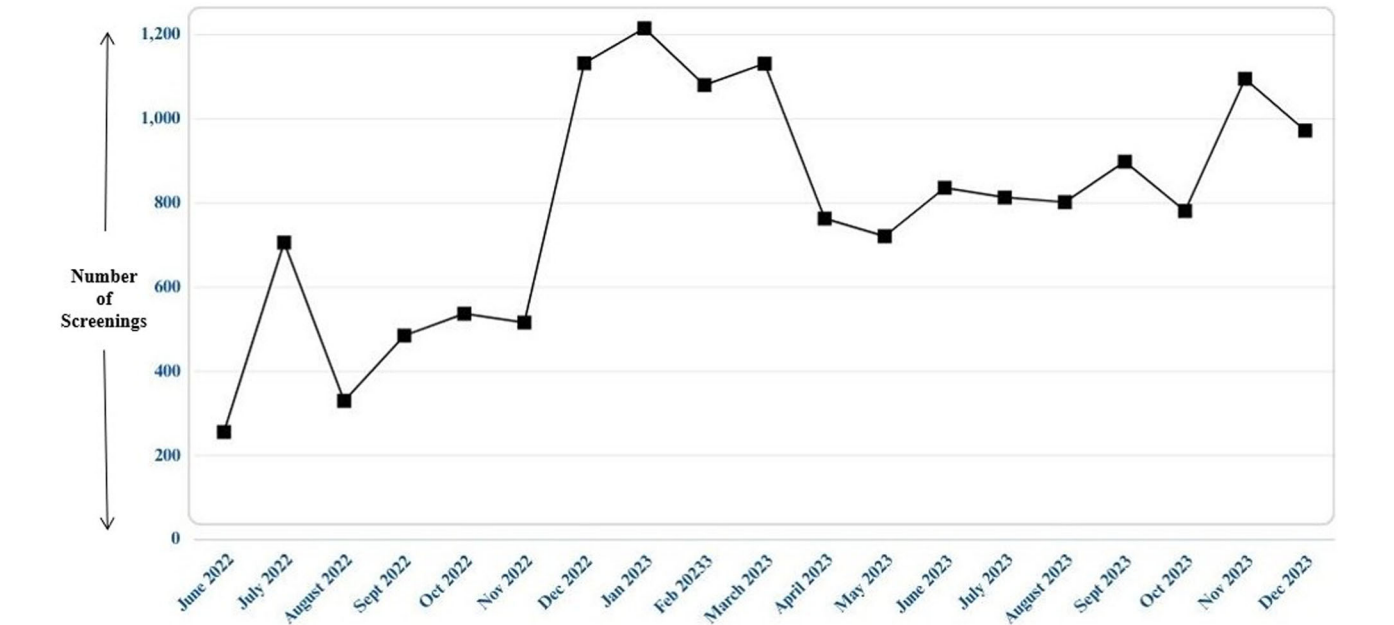


Fig. 2 | Number of screenings per month during the program duration. This figure depicts the monthly increase in the number of screenings performed across the duration of the program.

women were found to have benign lesions. The remaining cancers were detected in symptomatic women complaining of breast lumps.

Among the 27 women with histopathologically confirmed malignancies, lesion size information was available for 15 patients. The maximum lesion dimensions in this subset were as follows: 1 patient had a lesion ≤ 2 cm, 6 had lesions > 2 cm and ≤ 3 cm, 4 had lesions > 3 cm and ≤ 4 cm, 2 had lesions > 4 cm and ≤ 5 cm, and 2 had lesions > 5 cm. For the remaining 12 women, lesion size data were not available in their reports; however, 2 of these women were asymptomatic,

resulting in non-palpable lesions. At the time of writing this manuscript, treatment had been initiated for 18 out of the 27 histopathologically confirmed malignancies.

Diagnostic interval/interval to initiation of treatment

Following testing positive by Thermalytix, the PN’s ensured participant follow-up in a median of 6 days (IQR 4.5 days), leading to a diagnostic test confirmation in 21 days (IQR 10) and initiation of appropriate therapy within 30 days (IQR 14.5 days).

Table 1 | Demographics of the women screened

Characteristic	Number screened	Triaged positive	Number followed up	Cancer detected	PPV ^a
Age					
<30	878 [5.8%]	66 [7.5%]	39 [59.1%]	1	2.6%
30–40	6548 [43.5%]	210 [3.2%]	127 [60.5%]	5	3.9%
41–50	4374 [29.0%]	84 [1.9%]	43 [51.1%]	4	9.3%
51–60	2419 [16.1%]	46 [1.9%]	27 [56.7%]	6	22.2%
61–70	729 [4.8%]	43 [5.9%]	29 [67.4%]	10	34.4%
>71	121 [0.8%]	11 [9.1%]	3 [27.2%]	1	33.3%
Total	15,069	460 [3.1%]	268 [58.3%]	27	10.1%
Menopausal status					
Pre –menopause	9732 [64.6%]	330 [3.4%]	195 [59.1%]	8	4.1%
Post –menopause	5337 [35.4%]	130 [2.4%]	73 [56.2%]	19	26.0%
Symptomatic status					
Asymptomatic	10531 [69.9%]	108 [1.0%]	47 [43.5%]	2	4.3%
Symptomatic					
Complaint of breast swelling	1557 [34.3%]	304 [19.5%]	192 [63.2%]	25	13.0%
Breast pain	2934 [64.7%]	123 [4.2%]	80 [65.0%]	9	11.3%
Nipple discharge	204 [4.5%]	19 [9.3%]	16 [84.2%]	0	–
Non-specific complaints	412 [9.1%]	29 [7.0%]	19 [65.5%]	0	–
Post-mastectomy asymptomatic	106 [2.3%]	0	0	0	–
Post-mastectomy: symptomatic	23 [0.5%]	1 [4.3%]	1 [100%]	1	100%
Post-lumpectomy: asymptomatic	149 [3.3%]	5 [3.4%]	0	0	–
Post-lumpectomy symptomatic	186 [4.1%]	17 [9.1%]	13 [76.5%]	2	15.4%
Symptomatic	4538 [30.1%]	352 [7.8%]	221 [62.8%]	25	11.3%

^aPPV: cancer detected in women who screened positive and who underwent imaging follow-up.

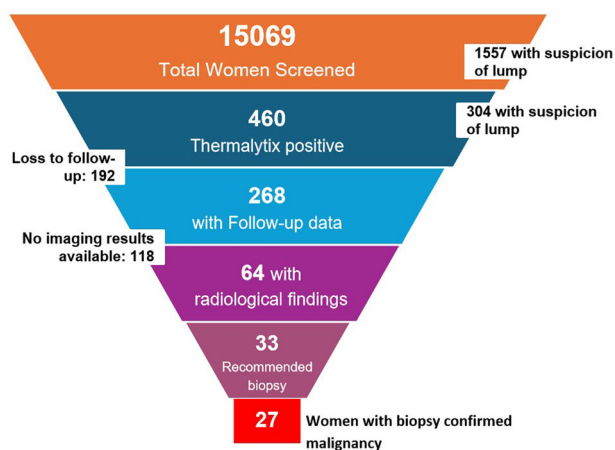


Fig. 3 | Summary of the outcomes of Thermalytix screening obtained in this program. This figure summarizes the key outcomes at different stages of the breast cancer screening program along with loss-to-follow-up.

Discussion

This program employed innovative digital tools such as online capacity building, screening with AI-based Thermalytix, and a digital referral pathway and showed a cancer detection rate (CDR) of 1.8 per 1000 screened with a recall of 3.1%, a Positive Predictive Value of recall (PPV1) of 10.1% (27/ 268), a Positive Predictive Value of biopsy recommended (PPV2) of 22% (33/150) and a Positive Predictive Value of biopsy performed (PPV3) of 81.81% (27/33) (<https://www.acr.org/-/media/ACR/Files/RADS/BI-RADS/FUOM-Basic-Audit.pdf>).

To compare the cancer detection rates with current reported data from India, we looked at incidence rates as per available cancer registries. The current standard of care breast screening program in India relies on NPCDCS recommended pathway of CBE-based screening provided through trained ASHA & ANM, with suspect cases being referred to Medical Officers at PHC for further evaluation and to the DH for imaging if deemed necessary. This screening pathway results in a nationwide AAI of 31.2/100,000 (Crude rate: 0.03%)⁵. In 2020, for the state of Punjab, CBE-based screening delivered an AAI ranging from 38–48/100,000 (Crude rate: 0.04–0.05%)⁴. As compared to these published reports, in our program, screening by Thermalytix delivered an AAI of 120.94/100,000 (Crude rate: 0.18%), which is a threefold improvement in cancer detection compared to CBE-based screening as per NPCDCS pathway.

The benefit of our digital-based program should also be considered alongside existing evidence from published CBE trials. Ngan et al.¹¹ reported an overview of systematic reviews of CBE, including 11 reviews, and reported that population coverage rates in screening programs based on CBE are suboptimal and the efficacy of CBE in downstaging disease or in reducing mortality remains unproven. This is likely due to a combination of factors. Firstly, the performance of CBE is entirely dependent on the skill of the examiner, with no standardization of technique, and no guidelines for optimal reporting³¹. Since the most sensitive technique has not been established, published clinical results report variable sensitivity of CBE of 40 to 69%, with a pooled sensitivity of 54.1%¹¹. Secondly, to be useful, effective CBE requires a trained and diligent HCW whose competence is trusted by the community. However, in most LMICs the HCW has many roles, is overburdened³² and in the face of competing healthcare priorities does not prioritize cancer screening. Of note, a recent Cochrane review found limited improvements in CBE detection rates even with the imparting of training to HCWs in CBE, again implying a need to strengthen any CBE-based screening service^{33,34}. Thirdly, sociocultural barriers represent the most

Table 2 | Comparison of the results of recent Clinical Breast Examination (CBE)-based screening experiences with the current program

Institution [reference]	TMC, Mumbai ⁹	RCC Trivandrum ¹⁰	JHPIEGO ³⁵	This study
Duration of study	1998–2019 [20 years]	2006–2019 [14 years]	2018–2020 [18 months]	2022–2023 [18 months]
Scale of screening	Multiple clusters in one district	Multiple clusters in one district	Two districts of Bihar/ Jharkhand	Entire state [23 districts]
Screening methodology	4 rounds of CBE and 5 rounds of active surveillance every two years.	3 rounds of CBE screening provided by trained health workers [HWs] at 3-year intervals	v-CHW,s implement health awareness/ community level CBE with referral to DH for further care	Line listing by existing v-CHW's followed by technology-based screening/ digital referral
Digital component	Absent	Absent	Absent	Both screening and follow-up employed digital tools
Scalability to state-level	Not attempted	Not attempted	Possible	Accomplished
Total screened	46554 ^A	38798 ^B	52248	15,074
Recall rate	1024[2.2%] ^A	2041[5.26%] ^B	1736 [3.3%]	460 [3.0%]
Follow-up completed	819[79.9%] ^A	data not adequate ^B	386 [22%]	268 [58.2%]
Screen-detected Cancer	30 ^A	38 ^B	24	27
Cancers detected of total number screened	0.06% ^A [30/46554]	0.09% ^B [38/38798]	0.04%	0.18%
Cancers detected in screened positive women	3.66% ^A [30/819]	Data not adequate ^B	6.2%	10.1%

^AThe data for these calculations were based on the findings of the fourth round of screening which is available in the data supplement of ref. 9.

^BThe source publication¹⁰ does not present findings for each of the 3 screening rounds separately. The numbers here reflect the mathematical average of all 3 rounds. For example, although 53418 women underwent at least one CBE, the average per round was 38798 [50366 in round 1 +34534 in round 2 +31495 in round 3 = 116395. 116395/3 is an average of 38798 women screened per round].

significant barrier to breast screening in most studies in both high-income and LMICs and are not lessened by CBE. Cultural norms of modesty such as embarrassment, shame on disrobing, husband's disapproval, and certain religious beliefs prevent women from accessing services requiring undressing in front of an HCW.

Since Ngan's review, three large trials of CBE-based screening have been reported from India^{9,10,35} and are summarized in Table 2. These trials were well-designed examples of improvements possible for CBE-based screening but focussed on localized populations without considering whether the interventions described in the methodology were suitable for scaling up to a state level, or beyond.

The results of the present study show a better cancer detection rate when compared with these landmark studies. Further, the current program was planned for and achieved state-wide rollout, heavily employing digital technologies to optimize all steps- be it online training, the use of a portable screening solution that allowed state-wide coverage on a rotational basis, or patient navigation using a novel digital referral pathway. The 3.1% test positivity with community screening in our program is similar to the other listed studies; however, as shown in row 11 of Table 2, the cancer detection rate observed in this program was at least two to fourfold that of other studies. Further, we were able to provide follow-up investigations to 58% of those screened positive, which is better than the follow-up obtained in the other studies. This is likely due to our utilization of focused patient navigation and a digital referral pathway.

Out of 1557 women who had a self-reported suspicion of breast lump, 1253 women tested negative with Thermalytix, indicating either normal or low-risk thermal abnormalities detected. In their previous publications, Thermalytix exhibited 98% negative predictive value for a test-negative, hence, for optimal utilization of our limited healthcare resources, we opted to focus further diagnostic evaluation only on the women who tested positive with Thermalytix ($n = 304$). Our digital-based program allowed suspect women to receive a diagnosis in 21 days and initiate treatment for breast cancer within 30 days. We believe that these performance metrics, when the program is expanded in the coming years, would lead to a downstaging of disease and a reduction of mortality.

Technologies utilizing artificial intelligence for image analysis generally alleviate shortages in skilled personnel and affordable systems and address the constraints of the availability of medical equipment¹³. Likewise, the AI-based automation of Thermalytix with its easy-to-use user interface allows the entire process of image capture to report generation to be performed even by a semi-skilled HCW at the point of care^{29,30}. The software mandates automated image quality checks thus ensuring standardized image capture. As compared to CBE which is a subjective and qualitative assessment, the use of Thermalytix also allows objective and quantitative analysis of any abnormality detected, greatly contributing to increased cancer detection rates. While existing research predominantly employs traditional machine learning and deep learning frameworks ref. 36,37 for the binary classification of thermal images as suspicious or non-suspicious, Thermalytix tool distinguishes itself through a novel radiomics-based artificial intelligence methodology ref. 19–30. This approach integrates a sophisticated workflow comprising multiple machine learning and deep learning algorithms, designed to generate interpretable outputs that support healthcare providers in making evidence-based clinical decisions.

Furthermore, the privacy awareness of Thermalytix addresses many barriers to breast screening. It is noncontact, non-invasive, radiation-free, and a non-breast compression test with high levels of patient satisfaction³⁸. The patient is partially disrobed inside the imaging cabin and the technician does not see or touch the patient at any time. The physical footprint required is small, and portable, thus allowing flexible deployment options even at remote health centres with minimal infrastructure.

Patient navigation has been shown to increase rates of screening participation, timely diagnosis, early institution of appropriate treatment, and treatment adherence³⁹. Although telephonic patient navigation is the most commonly used method, recent focus has been on improving the training of the PN workforce to improve outcomes, as we performed in this program. Recently, Tata Memorial Centre and Tata Institute of Social Sciences have launched an Advanced Diploma in Patient Navigation⁴⁰ which is a structured program to create the PN manpower pool required to form a bridge between patients and access to care, to fully realize the benefits of PN in cancer care. The availability of navigators in both clinical and community settings would expand their reach and impact.

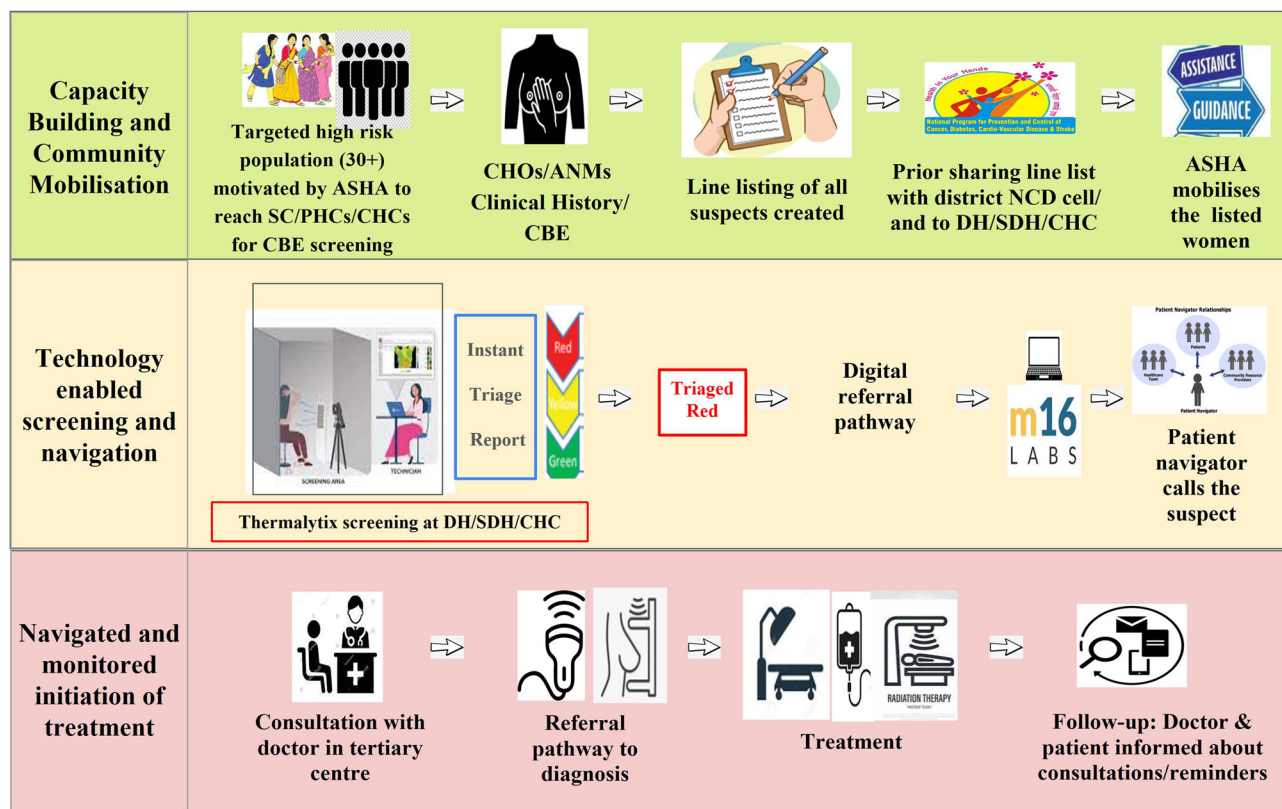


Fig. 4 | Illustration of the various components and flow of this program. This illustrates the key components of the program and the health facilities where the activities were conducted. SC sub-centres, PHC primary health centres, CHC

community health centres, CHO community health officers, NCD non-communicable disease, DH district hospital, SDH sub-divisional hospital, ANM Auxiliary Nurses and Midwife.

We acknowledge the limitations of this study. First, the lack of complete follow-up for all women who tested positive during screening prevents a full assessment of the program's impact. However, this challenge is common in India and is also a limitation in other trials, as noted in Table 2. It arises from the fragmented nature of India's healthcare system, where 80% of doctors and 57% of inpatient services operate in the private sector, resulting in 56.1% of individuals seeking private healthcare⁴¹. Additionally, nearly 42% of patients initially opt for traditional or complementary medicine after a cancer diagnosis, with a mean delay of 4 months (SD-68.75) before transitioning to allopathic care⁴².

Second, due to the nature of our study and the logistical challenges of conducting research in resource-limited settings, we couldn't collect detailed data on the proportion of women invited and deemed eligible for screening. This data would have provided further insights into the acceptability of the test. In future studies, we plan to address these limitations by conducting randomized controlled trials to compare the performance of Thermalytix against CBE and mammography, each in separate study arms. Further, large-scale studies conducted over a longer period are required to evaluate the post-deployment sustainability and maintenance of innovative AI tools for breast cancer screening at primary health centres.

This study presents an effective state-wide implementation of an innovative AI tool for breast screening at the population level in Punjab. Cancer screening achieved with low recall rates and acceptable PPV demonstrates the potential of Thermalytix for population-level breast screening. With the possibility of integration into existing primary healthcare systems, Thermalytix screening with patient navigation using a digital referral pathway may potentially alleviate the existing breast screening gap present around in resource-constrained countries and improve overall treatment outcomes. In the future, we plan to conduct large-scale randomized controlled trials with multiple study arms to compare the

effectiveness of Thermalytix, CBE, and mammography in terms of cancer detection rates and the cost-effectiveness of these techniques.

Methods

Overview

The program was multi-pronged and technology-based. The key elements of this program were (i) the capacity building of government HCWs to boost breast screening awareness in the community, (ii) a state-wide field study of the AI-based test for early detection of breast cancer, and (iii) a web-based digital referral system to ensure seamless patient navigation from screening and diagnosis, up to initiation of appropriate treatment for women identified with anomalies. DHFW would ensure prompt treatment services, thus ensuring a holistic approach to breast cancer management (Fig. 4). This study was conducted in accordance with the Declaration of Helsinki. The ethical approval (203/2029A01) for this research was granted by the Directorate of Health Services. The need for informed consent was waived by the ethics committee since all the data used in this study was acquired in the context of standard clinical care. Written consent was obtained from individuals involved in conducting the screening to publish images of the screening process.

The program aimed to cover all administrative blocks of the state of Punjab. Weeklong screening programs were conducted with 2 units of Thermalytix at 183 locations across 23 District Hospitals, 40 Sub-District Hospitals, and 26 Community Health Centers. Women aged 30 years and above were identified by village-level Community Health Workers (v-CHWs), primarily Accredited Social Health Activists (ASHA) and Auxiliary Nurses and Midwife (ANM) as requiring screening using Thermalytix. Additionally, at each screening location, women aged 18 years and above who attended breast awareness sessions and showed interest were allowed to undergo breast screening as an opportunistic screening. 15,000 women were to be recruited and screened over 18 months.

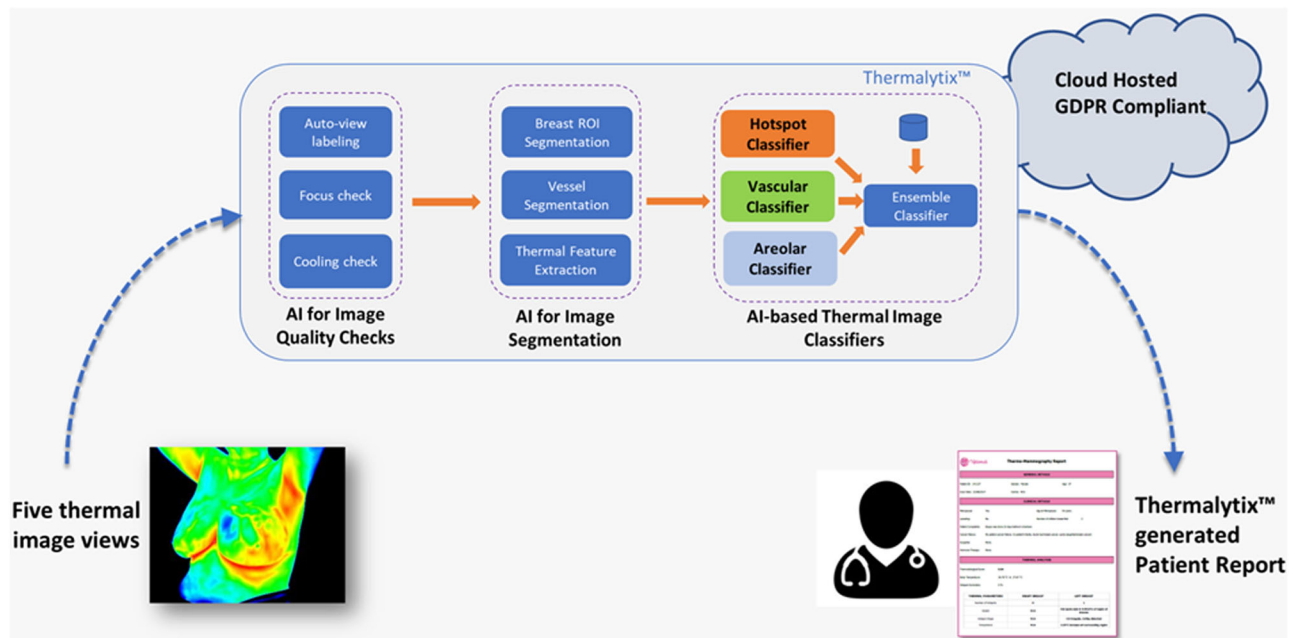


Fig. 5 | Key modules in Thermalytix. This illustrates the three distinct modules in Thermalytix, highlighting the transformation of raw thermal images into an interpretable report for clinician review.

Program Implementation

Phase 1: Capacity building and community mobilization. The following activities were conducted to boost breast screening awareness and facilitate the screening process.

- Regular educational sessions for all levels of v-CHWs on breast disease and high-risk breast symptoms, as well as training in CBE.
- Instructing v-CHWs on creating a line list of high-risk women requiring further screening and sharing this list with the District Non-Communicable Disease (NCD) cell to plan screening camps. We defined 'High-risk' women as those who were referred by v-CHWs based on clinical history and/or findings from CBE.
- Development of a communication strategy, including circulars, flyers, radio and newspaper advertising, by the Behavior Change Communication division of the National Health Mission Punjab.
- Regular liaison with the Community Process team (State ASHA Coordinator) to communicate to the ASHA on how to guide patients for screening.
- Continuing education for doctors on breast cancer epidemiology, signs & symptoms, clinical features, diagnosis, and treatment.
- Official communication to Civil Surgeons, District NCD cells, and District Collectors of each district for dissemination in the population, to motivate high-risk women to undergo breast screening.
- Sensitization meetings for all 23 district NCD cells, district Civil Surgeons, and Medical Superintendents to ensure the identification of adequate space in the facility to ensure patient privacy during screening.
- Government HCWs at all levels conducted breast awareness activities in their respective localities to mobilize appropriate women for breast screening.
- On the scheduled screening date, the concerned v-CHWs motivated and guided line-listed women to the screening location.

Over 18 months, a total of 22 training sessions on breast cancer risks, symptoms, and screening procedures were conducted—both in-person and online—for 23,008 v-CHWs, representing 81.4% of the existing v-CHWs. Each session lasted 90 min, with mandatory attendance required. The participating v-CHWs included 2848 out of 3916 Auxiliary Nurse Midwives

(ANMs) (89.1%), 16,929 out of 21,470 Accredited Social Health Activists (ASHAs) (79%), 611 out of 773 ASHA facilitators (79%), and 2617 out of 2823 community health officers (93%). Additionally, training was provided to 870 out of 1052 medical officers (83%).

The entire program was overseen by doctors at the Primary Health Centre (PHC), Sub-District Hospital (SDH), and District Hospital (DH), ensuring continuous monitoring. They were available to answer any questions raised by v-CHWs or participants and to counsel and motivate women who screened positive for further diagnostic imaging.

Phase 2: Breast cancer screening with novel AI-based technology, Thermalytix. Overview of Thermalytix AI modules: Thermalytix is a regulatory-approved software medical device designed to assess breast health by analyzing a series of five thermal scans. This cloud-based system processes thermal images to extract a detailed temperature map of the breast and utilizes a suite of advanced artificial intelligence (AI) modules ($n = 33$) to estimate the likelihood of breast malignancy. The AI workflow in Thermalytix is structured into three distinct modules:

1. AI for image quality: These ensure the acquisition of high-quality breast thermal images, critical for accurate analysis.
2. AI for image segmentation and radiomic feature extraction: This module employs automated image segmentation techniques to identify regions of interest (ROI) and extract novel vascular and thermal radiomic features.
3. AI-based thermal image classifiers: Leveraging the extracted radiomic features, this module applies machine learning classifiers to generate explainable AI outputs, culminating in a risk stratification score (referred to as the "B-Score"). This score provides actionable insights to guide healthcare providers in determining the next clinical steps.

Figure 5 illustrates the complete AI-driven pipeline, highlighting the transformation of raw thermal images into an interpretable report for clinician review.

AI-based image quality assessment. Thermalytix employs pre-trained deep learning and machine learning (ML) algorithms to perform automated image quality checks, addressing common imaging errors that

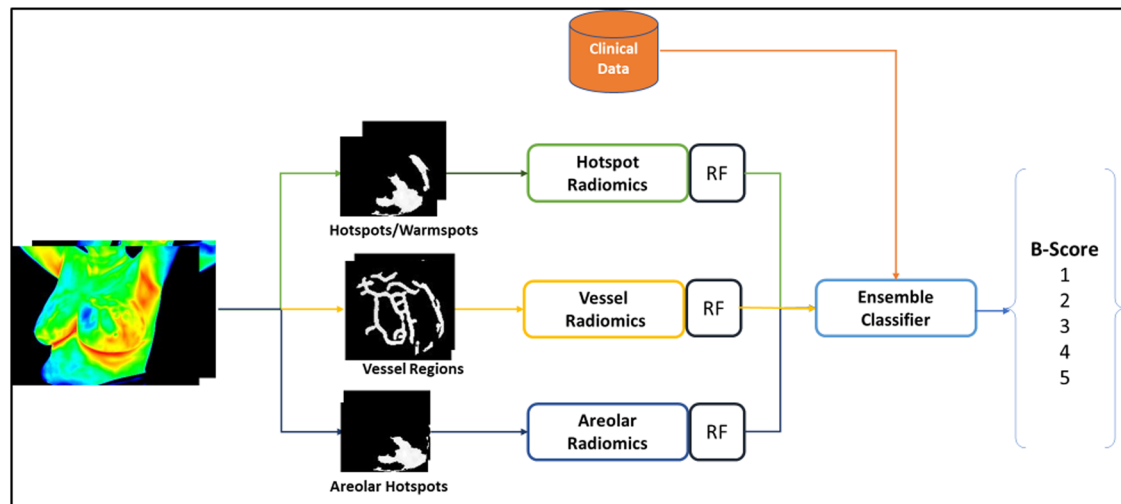


Fig. 6 | Overall AI workflow in Thermalytix. This illustrates the comprehensive AI workflow and innovative integration of explainable AI and radiomics in Thermalytix.

could otherwise compromise diagnostic accuracy. These include issues such as improper focus, inconsistent cooling, and view labeling errors, which are particularly prevalent when imaging is performed by minimally trained health workers in field settings. For instance, technicians may incorrectly label the right breast as the left or fail to capture all requisite views. The AI-powered view labeling module in Thermalytix demonstrated a remarkable accuracy of 99.53% in distinguishing between left and right breast views on an independent test dataset, ensuring high-quality imaging input for subsequent analyses³⁰.

AI for Region of Interest (ROI) segmentation. Thermal breast images typically encompass temperature distributions from the neck to the abdomen, necessitating precise segmentation of the breast region to exclude irrelevant areas. Thermalytix employs a V-Net architecture for this task^{29,30}. This architecture efficiently downsamples the input images through encoder stages to extract a latent representation and then upsamples the latent vector through decoder stages to generate high-resolution segmentation maps. Independent testing demonstrated that this segmentation module detects the breast region with a Dice index of 0.91 when compared to ground truth segmentation marked by an expert. This Dice index is also comparable to interobserver dice correlation, demonstrating the obtained accuracy and reliability with the segmentation module.

Thermal and vascular radiomics: Following segmentation, Thermalytix performs advanced analyses on the breast region to detect thermal constructs called hotspots, warm spots, and vascular regions. Hotspots and warm spots represent areas of elevated temperature and are identified through histogram-based thresholding techniques^{19,25,26}. A key aspect of determining hotspots and warm spots is the right estimation of a temperature threshold based on a decision function that is based on the temperature distribution on the breast and hence personalized to the patient. Furthermore, given that a malignant lesion often drives increased vascular activity to sustain its growth, a V-Net model is utilized to segment and detect vascular structures^{26,27}. This approach to vascularity detection is superior in comparison to other image processing techniques due to its improved vessel connectivity and computational time. Distinct radiomic features are then extracted from these subregions to capture distinct thermal patterns.

Thermalytix proposes novel hotspot radiomics and vascular radiomics to represent domain and data-based features that can be used to represent the unique characteristics of every patient's thermal scans, which can then be used to get an accurate classification. Hotspot radiomics quantify thermal asymmetries through features describing shape, size, symmetry, and

temperature variations, while vascular radiomics assess vessel morphology, including vessel count, branches, mean caliber, and symmetry. Thermal characteristics of the areolar region are also analyzed for symmetry.

Explainable AI for classification. The radiomic features derived from hotspots, vascular structures, and areolar regions serve as inputs for three independent ML classifiers, each producing a likelihood score for malignancy. These scores provide interpretability by linking suspicious findings to specific physiological abnormalities. Hotspot radiomics assess asymmetric inflammatory activity while vascular radiomics evaluate asymmetric vascular activity and areolar radiomics analyze milk duct-related irregularities. This novel explainable AI framework enhances clinician trust in the system by offering transparency into the rationale behind its predictions, aiding in clinical decision-making.

The likelihood scores from the three classifiers are further combined with patient clinical data to generate an ensemble score, termed the B-Score, which stratifies malignancy risk on a scale from 1 to 5: B-Scores of 1–2 (“Green”) suggest low risk, B-Score of 3 (“Yellow”) indicates moderate risk and B-Scores of 4–5 (“Red”) denote high malignancy likelihood.

This comprehensive AI workflow, illustrated in Fig. 6, exemplifies the innovative integration of explainable AI and radiomics in Thermalytix, setting a new standard for breast cancer screening.

Details of Real-world Implementation: All high-risk women aged 30 and above, as identified by v-CHW assessments, who expressed interest in the screening were recruited for the study. The decision to start screening at age 30 adheres to NPCDCS recommendations PN for initiating screening through the CBE pathway (<http://www.mohfw.gov.in/?q=Major-Programmes/non-communicable-diseases-injury-trauma/Non-Communicable-Disease-II/National-Programme-for-Prevention-and-Control-of-Cancer-Diabetes-Cardiovascular-diseases-and-Stroke-NPCDCS>). Additionally, women aged 18 and older who attended breast cancer awareness sessions and showed interest were allowed to undergo breast screening as part of an opportunistic screening process. Informed consent was obtained from all participants.

The breast cancer screening process using the AI-based Thermalytix technology involved the following steps. Participants entered a screening enclosure made from four curtains (see Fig. 7) to ensure privacy. They disrobed from the waist up and sat in front of an infrared camera (see Fig. 7) and a portable air-cooler for about five minutes to prepare for the test. The cooling period helped eliminate extraneous heat patterns and allowed the participants to reach thermal equilibrium with their surroundings. During this time, demographic information, breast-related complaints, clinical history, and personal and family cancer histories were collected. The technician monitored the cooling process, after which five thermal images were

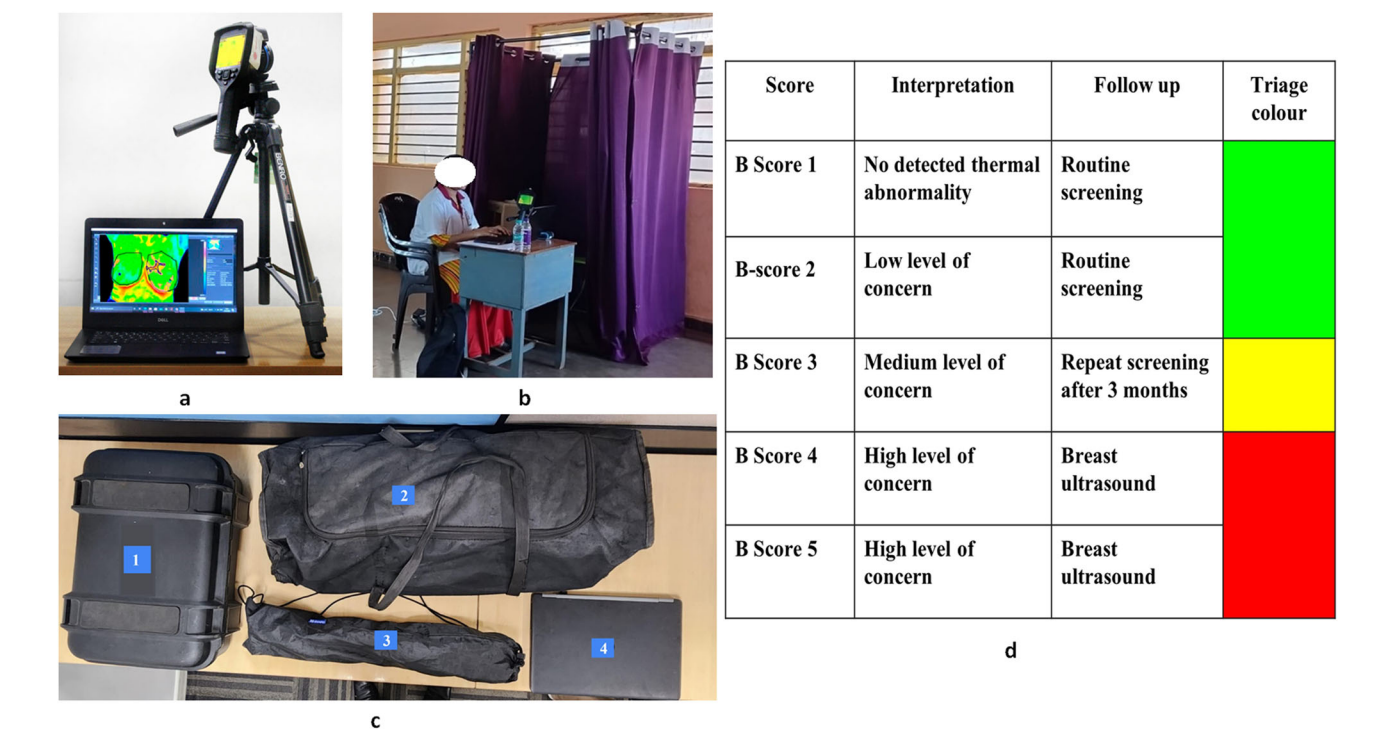


Fig. 7 | Breast cancer screening with Thermalytix. The breast cancer screening was conducted using AI-based Thermalytix. **a** Thermalytix device comprising a portable thermal sensor, a laptop, and cloud-based AI software. **b** Illustration of Thermalytix screening, where the participating woman is inside an enclosure created with cloth curtains providing privacy. **c** The complete portable Thermalytix setup with 1. camera case, 2. foldable screen structure with curtains, 3. camera tripod, 4. laptop. **d** Thermalytix scoring system.

taken from the neck to the abdomen in frontal, left-oblique, left-lateral, right-oblique, and right-lateral views.

These images were then uploaded to the cloud-based Thermalytix software, where pre-trained AI algorithms extracted key vascular and thermal radiomic features explained in Figs. 5 and 6 above. These radiomic features are automatically analyzed by machine learning classifiers to generate instant reports, the B-scores, which indicate the likelihood of malignancy. To ensure ease of comprehension for all educational levels of HCW, a color-coded Thermalytix output is automatically generated with a scale depicted in Fig. 7. Women categorized as “red” were advised to undergo further diagnostic evaluation. The entire process starting from patient preparation to report generation took approximately 12–15 min per person. A basic power supply of 110 W was required to operate the portable cooler and charge the laptop and camera. Figure 7 illustrates the complete portable Thermalytix setup, which includes the camera, foldable screens, curtains, camera tripod, and laptop.

The entire screening program was conducted using only two Thermalytix units, with one technician assigned per site. Five technicians (female, non-medical background, minimum of 12th-grade education) from the local community were recruited and trained to conduct the screenings. A native of Punjab was appointed as the program coordinator, along with patient navigators. The technicians underwent comprehensive training, including theoretical sessions and hands-on practice, with ongoing online and telephonic support until they achieved proficiency. Feedback during screening camps aided their learning, and periodic refresher training sessions were held. Any field issues were promptly addressed, and re-screenings were conducted, when necessary, with strict adherence to protocols throughout the program.

Phase 3: digital referral pathway. Roche India was involved in project planning, implementation & monitoring, specifically to strengthen the care pathway through a digital referral tracking system, with patient navigators (PN). Two PNs provided end-to-end counselling

to a woman who was triaged positive by Thermalytix to minimize loss-to-follow-up. They provided comprehensive assistance to screened-positive women on the processes needed to ensure final diagnosis, and the subsequent steps for appropriate cancer treatment.

PNs issued notifications and reminders through phone calls regarding further diagnostic consultations and logged the details in the digital referral application of the m16 platform. PNs were a one-stop source of appropriate and standardized information—regarding further imaging, staging, and treatment options, under the guidance of the registered medical practitioner. They also ensured that patient’s rights were recognized by maintaining confidentiality: women testing positive on screening could opt to keep their follow-up investigations kept confidential even to this program, if they so desired. The details of all detected cancer patients were linked to the state-wide PBCR to monitor their disease-free survival and mortality patterns.

The technicians and personnel involved in Thermalytix-based screening did not have access to any of the follow-up information of screened participants until the completion of the screening phase of the program.

Data availability

Statistical and demographic data that underlie the results reported in this article after de-identification will be shared upon request directly to the corresponding author. The data will be available beginning 3 months and ending 2 years following article publication. Researchers who provide a methodologically sound proposal should approach the corresponding author; to gain access, data requestors will need to sign a data access agreement. Ethical and legal implications of data sharing will be considered and the decision to share data will be based on the outcome of this review.

Code availability

The underlying code for this study and training/validation datasets is not publicly available for proprietary reasons.

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Author contributions

K.A., A.G., S.S., and H.K. conceived the study and methodology; K.A., A.T., A.S., and K.R. supervised the entire project and organized resources; K.A., A.G., H.K., and S.S. equally contributed to project administration, and writing

—review, and editing; S.S. and A.G. verified the underlying data reported in the manuscript. All authors have read and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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